

EXPERIMENT 2

2a) GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) SIGNALS USING MATLAB

OBJECTIVES:

- (i) To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.
- (ii) (ii) To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule.

OBJECTIVE 1: To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.

PROGRAM:

```
clc;

clear;

close all;

% Parameters

Am = 2;

Ac = 1;

fm = 5;

fc = 50;

beta = 5;

fs = 2000;

% Time vector

t = 0:1/fs:1;
```

```
% Message signal
```

```
m = Am * cos(2*pi*fm*t);
```

```
% Carrier signal
```

```
c = Ac * cos(2*pi*fc*t);
```

```
% FM signal
```

```
fm_sig = Ac * cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
```

```
% Plot signals
```

```
figure('Name','Frequency Modulation (FM)','NumberTitle','off');
```

```
subplot(3,1,1)
```

```
plot(t,m)
```

```
title('Message Signal')
```

```
xlabel('Time (s)')
```

```
ylabel('Amplitude')
```

```
grid on
```

```
subplot(3,1,2)
```

```
plot(t,c)
```

```
title('Carrier Signal')
```

```
xlabel('Time (s)')
```

```
ylabel('Amplitude')
```

```
grid on
```

```
subplot(3,1,3)
```

```
plot(t, fm_sig)
```

```
title('FM Signal')
```

```
xlabel('Time (s)')
```

```
ylabel('Amplitude')
```

```
grid on
```

```
% Display parameters
```

```
fprintf('Message Frequency = %d Hz\n', fm);
```

```
fprintf('Carrier Frequency = %d Hz\n', fc);
```

```
fprintf('Modulation Index ( $\beta$ ) = %.2f\n', beta);
```

OUTPUT:

```
1  clc;
2  clear;
3  close all;
4
5  % Parameters
6  Am = 2;
7  Ac = 2;
8  fm = 5;
9  fc = 50;
10 beta = 5;
11 fs = 2000;
12
13 % Time vector
14 t = 0:1/fs:1;
15
16 % Message signal
17 m = Am * cos(2*pi*fm*t);
18
19 % Carrier signal
20 c = Ac * cos(2*pi*fc*t);
21
22 % FM signal
23 fm_sig = Ac * cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
24
25 % Plot signals
26 figure('Name','Frequency Modulation (FM)','NumberTitle','off');
27
28 subplot(3,1,1)
```

Command Window

Message Frequency = 5 Hz
Carrier Frequency = 50 Hz
Modulation Index (β) = 5.00
>>



OBJECTIVE 2 : To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule.

PROGRAM:

```
clc;

clear;

close all;

%% Input Parameters

Am = 1.5;          % Message signal amplitude
fm = 1000;         % Message frequency (Hz)
Ac = 2;           % Carrier amplitude
fc = 20000;        % Carrier frequency (Hz)
kf = 2000;         % Frequency sensitivity (Hz/V)
fs = 200000;       % Sampling frequency (Hz)
T = 0.02;          % Simulation duration (s)

t = 0:1/fs:T;

%% Message Signal

m = Am * cos(2*pi*fm*t);

%% Frequency Deviation

delta_f = kf * Am;

%% Modulation Index

beta = delta_f / fm;

%% FM Signal
```

```
fm_signal = Ac * cos(2*pi*fc*t + beta * sin(2*pi*fm*t));
```

```
%% Plot Message and FM Signals
```

```
figure;
```

```
subplot(2,1,1)
```

```
plot(t,m,'LineWidth',1.5)
```

```
title('Message Signal')
```

```
xlabel('Time (s)')
```

```
ylabel('Amplitude')
```

```
grid on
```

```
subplot(2,1,2)
```

```
plot(t,fm_signal,'LineWidth',1.5)
```

```
title('Frequency Modulated Signal')
```

```
xlabel('Time (s)')
```

```
ylabel('Amplitude')
```

```
grid on
```

```
%% Spectrum Analysis
```

```
N = length(fm_signal);
```

```
FM_fft = fftshift(abs(fft(fm_signal))/N);
```

```
f = (-N/2:N/2-1)*(fs/N);
```

```
figure
```

```
plot(f,FM_fft,'LineWidth',1.5)
```

```
title('FM Signal Spectrum')
```

```
xlabel('Frequency (Hz)')
```

```
ylabel('Magnitude')
```

grid on

```
xlim([fc-10000 fc+10000])
```

%% Display Results

```
fprintf('\nFM SIGNAL PARAMETERS\n');
```

```
fprintf('-----\n');
```

```
fprintf('Carrier Frequency    = %.0f Hz\n',fc);
```

```
fprintf('Message Frequency      = %.0f Hz\n',fm);
```

```
fprintf('Message Amplitude      = %.2f V\n',Am);
```

```
fprintf('Frequency Sensitivity  = %.0f Hz/V\n',kf);
```

```
fprintf('\nCALCULATED VALUES\n');
```

```
fprintf('-----\n');
```

```
fprintf('Frequency Deviation ( $\Delta f$ ) = %.0f Hz\n',delta_f);
```

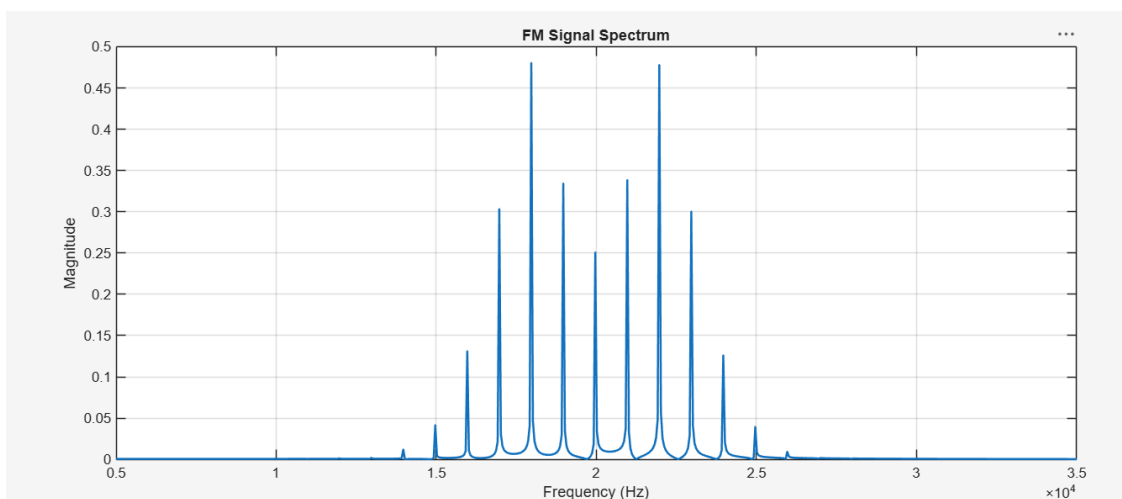
```
fprintf('Modulation Index ( $\beta$ )    = %.2f\n',beta);
```

%% Carson's Rule Bandwidth

```
BW = 2 * (delta_f + fm);
```

```
fprintf('Bandwidth (Carson Rule) = %.0f Hz\n',BW);
```

OUTPUT:



FM SIGNAL PARAMETERS

Carrier Frequency = 20000 Hz
Message Frequency = 1000 Hz
Message Amplitude = 1.50 V
Frequency Sensitivity = 2000 Hz/V

CALCULATED VALUES

Frequency Deviation (Δf) = 3000 Hz
Modulation Index (β) = 3.00
Bandwidth (Carson Rule) = 8000 Hz

2b) GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) SIGNALS USING MATLAB

OBJECTIVE : To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule

PROGRAM:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Message Signal Parameters
```

```
Am = 2; % Message amplitude (V)
```

```
fm = 500; % Message frequency (Hz)
```

```
%% Carrier Signal Parameters
```

```
Ac = 2; % Carrier amplitude (V)
```

```
fc = 10000; % Carrier frequency (Hz)
```

```
%% PM Parameters
```

```

kp = 2;      % Phase sensitivity (rad/V)
fs = 100000; % Sampling frequency (Hz)
T = 0.01;    % Simulation duration (s)

%%% Time Vector
t = 0:1/fs:T;

%%% Message Signal
m = Am*cos(2*pi*fm*t);

%%% Phase Modulation Index
beta = kp * Am;

%%% PM Signal
pm_signal = Ac*cos(2*pi*fc*t + kp*m);

%%% Display Results
fprintf('OUTPUT\n');
fprintf('-----\n');
fprintf('Message Frequency (fm) = %.0f Hz\n',fm);
fprintf('Message Amplitude (Am) = %.2f V\n',Am);
fprintf('Phase Sensitivity (kp) = %.2f rad/V\n',kp);

fprintf('\nCALCULATED VALUES\n');
fprintf('-----\n');
fprintf('Phase Deviation = %.2f rad\n',beta);
fprintf('Modulation Index (Beta) = %.2f\n',beta);

%%% Plotting

```

```
figure;
```

```
subplot(2,1,1)
```

```
plot(t,m,'LineWidth',1.5)
```

```
grid on
```

```
title('Message Signal')
```

```
xlabel('Time (s)')
```

```
ylabel('Amplitude (V)')
```

```
subplot(2,1,2)
```

```
plot(t,pm_signal,'LineWidth',1)
```

```
grid on
```

```
title('PM Signal')
```

```
xlabel('Time (s)')
```

```
ylabel('Amplitude (V)')
```

```
xlim([0 0.01])
```

OUTPUT:

OUTPUT

```
-----  
Message Frequency (fm) = 500 Hz  
Message Amplitude (Am) = 2.00 V  
Phase Sensitivity (kp) = 2.00 rad/V
```

CALCULATED VALUES

```
-----  
Phase Deviation = 4.00 rad  
Modulation Index (Beta) = 4.00  
>>
```

